Lecture 4

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Assignment 3: please hand it in now
Assignment 4: please take a paper copy

Unit 4: Discrete Probability Distributions[1]

Overview

In this lecture, we explore four essential discrete probability distributions that frequently appear in computing applications. Each distribution models a specific type of random process commonly encountered in software systems, networks, and algorithms.

Today's Topics:

1. Binomial Distribution – fixed trials, success/failure outcomes
2. Hypergeometric Distribution – sampling without replacement
3. Geometric Distribution – trials until first success
4. Poisson Distribution – events over time/space

1. Binomial Distribution

Definition and Properties

The binomial distribution models the number of successes in a fixed number of independent trials, where each trial has only two possible outcomes (success or failure) and the probability of success remains constant.

Notation: $X \sim Binomial(n, p)$ $X \sim Binom(n, p)$

$\sim$ = follows distribution so and so

Where:

- n = number of trials
- p = probability of success on each trial (remains constant)
- q = probability of failure on each trial $q = 1 - p$
- X = number of successes $X = \{0, 1, 2, \dots, 10\}$

Probability Mass Function (PMF):

$$\boxed{P(X = k) = C(n, k) \times p^k \times (q)^{n-k} \quad k = 0, 1, 2, \dots, n}$$

Mean and Variance:

- $Mean: \mu = np$ ✓
- $Variance: \sigma^2 = npq$ ✓

$E[X] = \sum x \cdot p(x) = np$

$\sum x^2 \cdot p(x) - \mu^2 = npq$

[1] [1] I have used Claude Sonnet 4.5 to edit, structure and generate some of the examples in these lecture notes

$$\frac{\cancel{100!}}{\cancel{98!}\,2!} = \frac{100\times 99}{2}$$

CS Examples:

Example 1: A network router sends 100 packets. Each packet has a 95% probability of successful transmission (no corruption or loss). What is the probability that exactly 98 packets are transmitted successfully?

$n = 100$

$p = 0.95$

$q = 0.05$

$$P(X=98) = C(100,98)(0.95)^{98}(0.05)^{2} = 0.08118$$

$\simeq 8\%$

```
> dbinom(98, 100, 0.95)
[1] 0.08118177
```

Example 2: A software deployment has a 98% success rate. If you deploy to 50 servers, what is the probability that:

$n = 50$ $\quad 0 \le x \le 50$

$p = 0.98$

$q = 0.02$

a) Exactly 45 deployment succeed.

b) At least 45 deployments succeed.

c) At most 45 deployments succeed.

a) $P(x=45) = C(50,45)(0.98)^{45}(0.02)^{5} = 0.0027$

```
> dbinom(45, 50, 0.98)
[1] 0.002731525
```

b) $P(x \geq 45) = \sum_{x=45}^{50} P(x)$

$P(x \le 45) = \text{pbinom}$

$= 1 - P(x \le 44)$

```
Either: 1-pbinom(44,50,0.98)
[1] 0.9995218

Or: pbinom(44, 50,0.98, lower.tail = 0)
[1] 0.9995218
```

c) $P(x \le 45) = \sum_{x=0}^{45} P(x) =$

```
> pbinom(45,50,0.98)
[1] 0.003209742
```

2. Hypergeometric Distribution

Definition and Properties

The hypergeometric distribution models the number of successes when sampling without replacement from a finite population containing both successes and failures. Unlike the binomial distribution, probabilities change as items are sampled.

$Notation: X \sim Hypergeometric(N, K, n)$ $\quad n \leq N$

Where:

- N = population size
- K = number of success states in population
- n = number of draws (sample size)
- X = number of observed successes in sample

Probability Mass Function (PMF):

$$P(X = k) = \frac{[C(K,k) \times C(N-K, n-k)]}{C(N,n)}$$

Mean and Variance:

$$Mean: \mu = n \times \left(\frac{K}{N}\right)$$

$$Variance: \sigma^2 = n \times \left(\frac{K}{N}\right) \times \left(1 - \frac{K}{N}\right) \times \left[\frac{N-n}{N-1}\right]$$

$N = 50$ $\quad K = 8$ bad

$n = 10$ $\quad 42$ good

CS Examples: Software Testing

A software module contains 50 functions, of which 8 have bugs. A test suite randomly selects 10 functions to test.

a. What is the probability that exactly 2 buggy functions are tested?

b. What is the probability that less than 2 buggy functions are tested?

a) $P(x=2)$ $\quad \frac{C(8,2) \times C(42,8)}{C(50,10)} = 32.17\%$

$x \quad K \quad N-K \quad n$

```
> dhyper(2,8,42,10)= 0.3217247
```

b) $P(x=0) + P(x=1)$ $\quad \frac{C(42,10)\,C(8,0)}{C(50,10)} + \frac{C(42,9) \times C(8,1)}{C(50,10)}$

$= 0.14324 + 0.3473 = 0.4905 \simeq \boxed{49.1\%}$

```
phyper(1,8,42,10)
[1] 0.4905024
```

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Another example: Security Vulnerability Audit

A software company has deployed 18 microservices in production. A security audit reveals that 8 of these microservices have critical vulnerabilities. The security team randomly selects 6 microservices for immediate patching. What is the probability that:

a) they will identify at least 1 microservice with a critical vulnerability?

b) They will identify 5 microservices with a critical vulnerability?

c) What is the expected number of microservices with a critical vulnerability from this sample? Is 5 a typical number of microservices with a critical vulnerability?

Small population (N=18), sampling without replacement
Good: 10
Bad: 8 = success
Sample size: 6

a) $P(x \geq 1) = 1 - P(x = 0) = 1 - \frac{C(10,6) \times C(8,0)}{C(18,6)} = 1 - 0.0113 = 0.9886877$

```
> 1-dhyper(0,8,10,6)= 0.9886878
```

b) $P(x = 5) = \frac{C(8,5) \times C(10,1)}{C(18,6)} = 0.030165912$

```
dhyper(5,8,10,6)= 0.03016591
```

In a sample of size

$\boxed{2.7 \pm 1 = \text{Average number of systems with issues.}}$

$\mu = 6 \times \frac{8}{18} = 2.7$

$\sigma^2 = 6 \times \frac{8}{18} \times \left(1 - \frac{8}{18}\right)\left(\frac{18-6}{18-1}\right) = 1.04575$

$\sigma = 1.02$

for $x = 5$: $Z_{score} = \frac{5 - 2.7}{1} = 2.3$ It is not typical.

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$X = 4$

3. Geometric Distribution

X = trials till the first success

4th trial = first success

Definition and Properties

The geometric distribution models the number of trials needed to get the first success in a sequence of independent Bernoulli trials. This is the "memoryless" discrete distribution.

Notation: X ~ Geometric(p)

Y = fails before the first success

3 fails — pass on 4th

$Y = 3$

Where:

- p = probability of success on each trial
- q = probability of failure on each trial
- X = number of trials <u>until</u> first success (including the success)

Note: *R uses the alternative parameterization where X counts failures <u>before</u> the first success. Be careful with interpretation!*

Probability Mass Function (PMF):

$$P(X = k) = q^{k-1} \times p$$
$$for\ k = 1, 2, 3, \ldots$$

Mean and Variance:

$$Mean: \mu = \frac{1}{p}$$
$$Variance: \sigma^2 = \frac{q}{p^2}$$

Example: Suppose a tech-support help line is occupied 75% of the time. Compute the probability that you will have to dial 5 times to gain access?

success: talking to someone

failure: not gaining access.

$P(X=5) = (0.75)(0.75)(0.75)(0.75)(0.25)$

$= (0.75)^4 (0.25) = 0.0791 = 7.91\%$

```
dgeom(4,0.25)
= 0.07910156
```

0.25 = prob success

4 = number of fails.

Example: A randomized algorithm has a 20% chance of finding the solution on each attempt. Compute the probability that you will have to run the algorithm 6 times to find the solution. What is the expected number and standard deviation of attempts needed?

$P(X=6) = (0.8)^5 (0.20) \simeq 6.6\%$

summary: 5 ± 4.5

$\mu = \frac{1}{p} = \frac{1}{0.20} = 5$

$var = \frac{q}{p^2} = \frac{0.8}{0.2^2} = 20$ $\quad \sigma = \sqrt{20} \simeq 4.5$

4. Poisson Distribution

Definition and Properties

The Poisson distribution models the number of events occurring in a fixed interval of time or space, when these events happen independently at a constant average rate. It's crucial for modeling arrival processes and rare events.

Notation: $\boldsymbol{X \sim Poisson(\lambda)}$

Where:

- λ = average rate (events per interval)
- X = number of events in the interval

Probability Mass Function (PMF):

$$P(X = k) = \frac{(\lambda^k \times e^{-\lambda})}{k!}$$
$$for\ k = 0, 1, 2, 3, \ldots$$

Mean and Variance:

- $Mean: \mu = \lambda$
- $Variance: \sigma^2 = \lambda$

Note: Mean equals variance is a distinctive property of Poisson distribution.

Computing Example: Server Request Arrivals

Scenario: A web server receives an average of 12 requests per minute. Assuming requests arrive independently, what is the probability of receiving exactly 15 requests in a given minute?

unit of time: minute

$\mu = 12 \quad \frac{\text{guests}}{\text{minute}}$

$P(x = 15) =$

```
dpois(15, 12)
[1] 0.07239112
```

Part b) At least 15 guests in a given minute? $P(\geq 15) = 1 - P(x \leq 14) = 1 - ppois(14, 12)$ =0.2279755

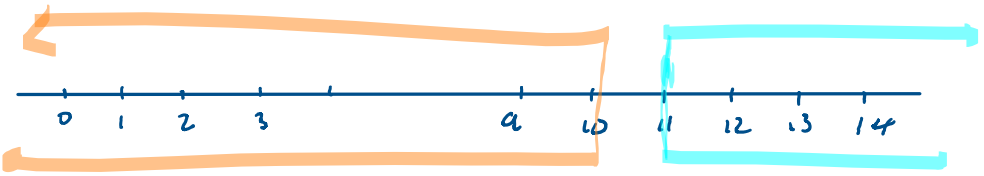
Additional Example: System Failures

Scenario: A data center experiences server failures at an average rate of 0.5 failures per day. What is the probability of having no failures in a given day?

$\mu = \lambda = 0.5 \quad \frac{\text{fail}}{\text{day}}$

$P(x = 0) = dpois(0, 0.5) =$

$0! = 1$



Example: A system receives an average of 8 security alerts per hour. What is the probability of receiving more than 10 alerts in a given hour?

$\mu = 8 \ \frac{\text{alerts}}{\text{hour}}$ $\quad P(x>10) = P(x \geq 11) = 1 - P(x \leq 10)$

$= 1 - \text{ppois}(10, 8)$.

Receive more than 10 alerts in a 3-hour time interval? $\mu = 24 \ \frac{\text{alerts}}{\text{3 hours}} = 1 - \text{ppois}(10, 24)$

Quick Reference Comparison

$=5$ $\quad \leq 5$

Distribution	What it Models	Parameters	Mean	R Functions
Binomial	Fixed trials, count successes	n, p	np	dbinom, pbinom
Hypergeometric	Sample without replacement	N, K, n	$n\left(\frac{K}{N}\right)$	dhyper, phyper
Geometric	Trials until first success	p	$\frac{1}{p}$	dgeom, pgeom dgeom(3)
Poisson	Events over interval	μ	μ	dpois, ppois

Key Takeaways

Choosing the Right Distribution:

- **Binomial:** Fixed number of trials, want to count successes
- **Hypergeometric:** Sampling from finite population without replacement
- **Geometric:** Want to know how many trials until first success
- **Poisson:** Events occurring over time/space at constant rate

R Functions Pattern:

- d-function (e.g., dbinom): probability mass function $P(X = k)$
- p-function (e.g., pbinom): cumulative probability $P(X \leq k)$
- r-function (e.g., rbinom): generate random samples

Important Notes:

- R's geometric distribution counts *failures* before first success, not total trials
- Use lower.tail = FALSE for $P(X > k)$ calculations
- Always verify your parameters match the problem's context